

## AI TRAINING – ASSIGNMENT 01

### AI Fundamentals: Explore, Understand & Explain

*Self-Exploration Assignment | Beginner Level*

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| <b>Topic Area</b>     | AI overview, AI vs ML vs DL vs GenAI, use cases, and career map       |
| <b>Mode</b>           | Self-exploration + research + practical thinking                      |
| <b>Suggested Time</b> | 2–3 hours                                                             |
| <b>Submission</b>     | One document / PDF containing your answers, examples, and reflections |

#### 1. What is this assignment about?

This is not a memorization assignment. Your goal is to explore AI yourself, understand the basic concepts, and explain them in simple words. You are expected to search, watch short videos if needed, use AI tools for clarification, and compare information from multiple sources.

Important: Do not simply copy an AI-generated answer. Use AI as a learning assistant, then rewrite the ideas in your own words and add your own examples.

#### 2. Learning Goals

- Understand what Artificial Intelligence means at a high level.
- Clearly distinguish AI, Machine Learning, Deep Learning, and Generative AI.
- Identify real-world AI use cases across different industries.
- Understand the major career paths available in AI and related technology roles.
- Explain the concepts in your own words instead of relying on textbook definitions.
- Apply the concepts to a practical real-world example.
- Identify common misunderstandings or errors and improve your explanation.

#### 3. Your Mission

Imagine that a friend who knows nothing about AI asks you: “What exactly is AI, what is the difference between AI, ML, DL and GenAI, where is AI used, and what can I become if I learn AI?”

Your job is to create a beginner-friendly answer that your friend can understand in 10 minutes.

#### 4. Step 1 – Explore: What is AI?

Do your own research and answer the following questions:

1. What is Artificial Intelligence in your own words?  
Answer: Artificial intelligence is a field of computer science and engineering .it creates machines or software that can perform tasks like learning, problem-solving, and understanding like a human.
2. What makes a system 'intelligent'?

Answer : Understanding the problem, learning from the given dataset, and making suitable predictions make the system intelligent.

3. Give at least 5 examples of AI that you may interact with in daily life.

Answer: AI examples are AI chatbots like GPTs, YouTube recommendations, Face Recognition Systems, Disease Recognition Systems in agriculture, New content generation

4. Choose one AI example and explain what the system is doing.

Answer: I choose FRS as an example of AI. When a face is captured by the camera, it analyzes the face, recognizes the person, and marks their attendance in the database.

5. Is every software application an AI application? Explain why or why not.

Answer: No, not every software application is an AI application because some applications work on predefined rules. For example, disease analysis in hospitals.

6. What are 2 things AI can do well and 2 things AI may struggle with?

Answer: AI is good at analyzing large amounts of data and performing repetitive tasks quickly. AI is not good at common sense and understanding human emotions.

Research rule: Use at least 3 learning sources. These can be articles, videos, official documentation, or educational websites.

Tip: Try searching for simple explanations first. If two sources disagree, investigate why.

## 5. Step 2 – The Big Picture: AI vs ML vs DL vs GenAI

This is the most important part of the assignment. Do not memorize definitions. Build a mental model.

First, complete this comparison table in your own words:

| Concept | What is it?                                                                                                                                                               | How does it learn/work?                                                                                                                                                                                 | Simple example                                                                                                                                                                                                                                                                     | Relationship                                                                                          | Your analogy                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------|
| AI      | AI is a field of computer science and engineering. It creates machines or software that can perform tasks like learning, problem-solving, and understanding like a human. | It can learn from the dataset which we provide, and its workflow is: it takes the inputs from the user, then preprocesses them and gives them to the ML model, and that ML model gives the predictions. | When a farmer scans the leaf of a crop, the AI can take it as an input, then clean the unnecessary data in the photo and give it to the ML model. The ML model is already pretrained with some related crop data. Then, it analyzes the disease in that leaf by using DL and gives | AI is a big field. In that field, ML and DL are the subfields used to improve the output predictions. | AI is like a school where machines learn |

|                       |                                                                                                                                              |                                                                                                                                                                                                         |                                                                                                                                                                                                                |                                                                                                 |                                                                                                             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|                       |                                                                                                                                              |                                                                                                                                                                                                         | the output.                                                                                                                                                                                                    |                                                                                                 |                                                                                                             |
| Machine Learning (ML) | ML is a machine that learns from the dataset and gives predictions.                                                                          | It takes the inputs from the users, and the model is already trained from the given dataset by using the algorithms of classification, regression, and clustering, and gives the output as predictions. | For example, when an email comes to the inbox, the ML model tells us whether the email is spam or not spam by using classification algorithms like Decision Tree, Random Forest, SVM, and Logistic Regression. | ML is a subfield of AI and DL that helps in critical situations like image and voice analysis.  | ML is like a student in a school that can learn from the dataset. This dataset is like a teacher in school. |
| Deep Learning (DL)    | DL is a subfield of ML. It has neural networks that help to analyze images, audio, and videos step by step, and it gives patterns as output. | DL takes the inputs and analyzes them layer by layer. These layers have neural networks that make them into patterns and give them to us as outputs.                                                    | For example, DL is used in image processing. If you give a group photo, in that photo there are cats, dogs, and monkeys. The DL model classifies how many dogs, cats, and monkeys are in that photo.           | It is a subfield of ML. It helps with critical analysis in AI and ML and is also used in GenAI. | DL is like a student learning deeply in the textbooks .                                                     |
| Generative AI (GenAI) | GenAI is also a subfield of AI that can create new data.                                                                                     | It uses large amounts of data from datasets. Based on that, it creates new data according to the prompt.                                                                                                | When a user gives a prompt to create a plan for 3 days according to the user's rules, it can generate the plan.                                                                                                | GenAI is related to AI, ML, and DL to create new things.                                        | Students like GenAI because they can create new projects.                                                   |

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Now answer these questions:

1. Is ML a part of AI? Explain with a simple example.

Answers: Yes, ML is a part of AI because AI can use ML to give the output by using ML models.

2. Is Deep Learning a type of Machine Learning? Explain.

Answers: Yes, DL is a type of ML. In DL, it has neural networks to give the patterns.

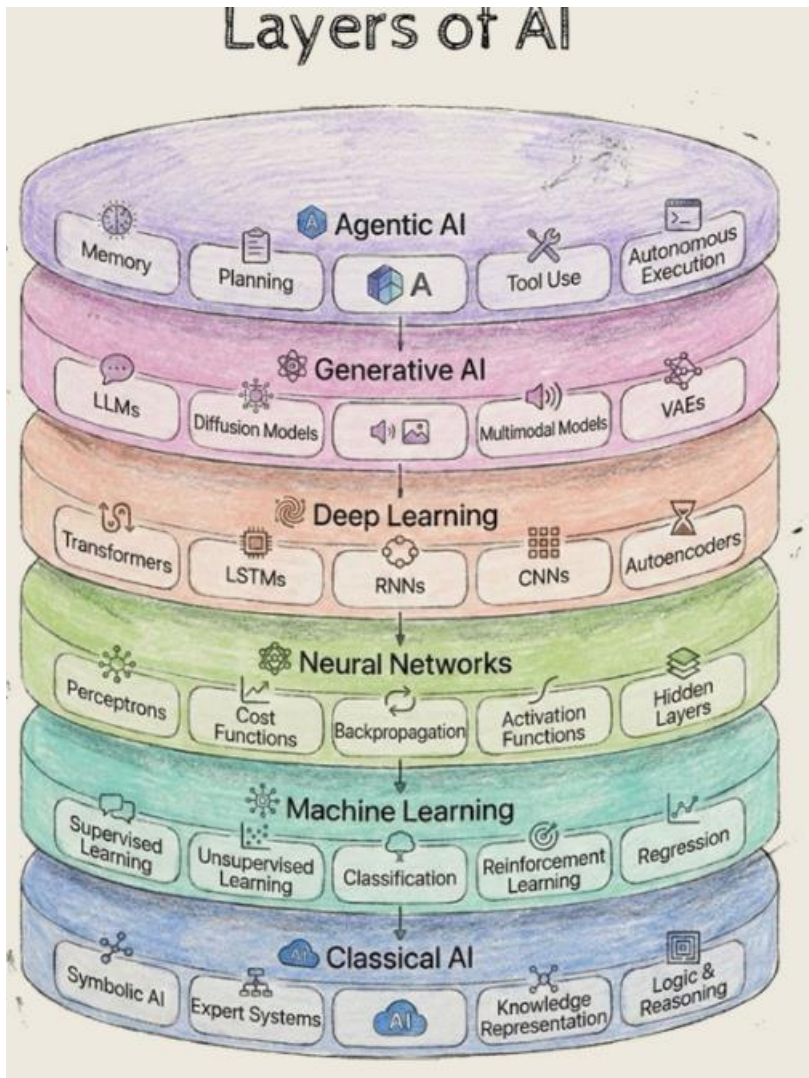
3. Is Generative AI the same thing as AI? Why?

Answers: No, GenAI is not the same thing as AI because GenAI can create new things, but AI can give answers that already exist in the data.

4. Can an AI system exist without Machine Learning? Research and explain one example.

Answer: Yes, AI systems can exist without ML. Rule-based AI can work with predefined rules. For example, if you give the input as "Hi", it responds with "Hello".

5. Draw a simple diagram showing the relationship between AI, ML, DL and GenAI.



6. Create one real-world analogy to explain all four concepts together.

Answer: In a water pipeline system, sensors collect data like water pressure and flow rate, AI identifies problems, and ML predicts problems like water leakage.

DL analyzes complex patterns in the sensor data, and GenAI generates a report or explanation about the detected problem.

### 6. Step 3 – Use Case Hunt

Find AI applications in different industries. Do not use only ChatGPT examples.

| Industry          | AI use case                      | AI / ML / DL / GenAI?                                                                                                   | Who benefits?                                                                                 | What could go wrong?                                                                                                        |
|-------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Banking / Finance | Customer care.<br>Loan sanction. | AI takes the customer's doubts or problems and gives answers. ML classifies which category the problem belongs to, like | The bank workers and customers can benefit by using them to save time and get fast responses. | Sometimes, the analysis may be wrong because the person who took the loan may suddenly lose their job. How can they pay the |

|                     |                                                                                                          |                                                                                                                                                                                                                                                  |                                                                                                                                           |                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |                                                                                                          | money belonging to savings or business. DL analyzes the problem given by the customer, and GenAI creates a new answer for the customer. ML takes the previous data of the person to check whether the person pays the loan money on time or not. |                                                                                                                                           | loan?                                                                                                                                                                                            |
| Healthcare          | The report can analyze the problem and give a little suggestion for first aid.                           | ML, AI, and DL can be used here to analyze the reports, and GenAI can help give suggestions.                                                                                                                                                     | Doctors and nurses can benefit by saving time, and patients can know the results faster. Doctors can also check more patients.            | Sometimes, the analysis of the X-rays and lab reports may be wrong.                                                                                                                              |
| Education           | personal tutor                                                                                           | All four—AI, ML, DL, and GenAI—can be used to explain the concept and create new examples.                                                                                                                                                       | Students and lecturers can learn new things.                                                                                              | Students can learn the concept blindly, and then they think it is correct sometime the response goes wrong                                                                                       |
| Retail / E-commerce | Product recommendations                                                                                  | AI, ML, and DL can be used to recommend projects in another app to improve sales.                                                                                                                                                                | Customers can view the best products, and salespersons can sell their products faster.                                                    | The creator creates content for their viewers, but the information may be wrong.                                                                                                                 |
| Manufacturing       | Automate repetitive tasks                                                                                | AI, ML, and DL can automate repetitive tasks using fixed rules or by learning from data, including complex images and speech. GenAI can repeatedly create new content such as text, images, audio, or code.                                      | Owners have less work, so they do not need to pay more money to workers.                                                                  | In manufacturing, if we use chatbots, sometimes they can give wrong information or wrong instructions. They may also not understand the problem correctly, which can cause mistakes in the work. |
| Agriculture         | Crop disease analysis and suggestions, the crops that are suitable for the soil, and weather prediction. | AI, ML, and DL are used to predict diseases, and GenAI can create text or prompts to answer farmers' questions with suitable answers.                                                                                                            | Farmers can benefit by knowing which crop diseases they have, what precautions they should take, and which crops are good for their soil. | Farmers may face problems if they trust AI completely because it may give wrong analysis. For example, it may predict good weather, but at that time, it may rain heavily.                       |

|                       |                                                                     |                                                                                                                                                                     |                                                                                                                                    |                                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transportation        | Prediction of traffic and route optimization to save time and cost. | ML and AI are used to analyze previous traffic data to predict the present traffic and tell us which route is fast and easy to reach the destination.               | Daily workers and traffic police can benefit by reducing traffic.                                                                  | Drivers may face problems if the AI and ML models learn from yesterday's data, but today some policies are made on this route and the traffic becomes heavy. Then, the prediction may go wrong. |
| Entertainment / Media | Create new content or explain the concept using an image.           | GenAI is used to generate new content, and AI, ML, and DL are used to understand the user's prompt. This prompt may be in the form of text, image, audio, or video. | Content creators and viewers can benefit. The creators earn money, and the viewers understand the concept faster and more clearly. | Content creators may create wrong information, and viewers may think that the information is true.                                                                                              |

Challenge: Find one AI use case that you think is surprisingly useful and explain why.

#### 7. Step 4 – Mini Practical Exercise: Design an AI Solution

Choose ONE problem from your college, personal life, local community, or a business you know. Design a simple AI-based solution. You do not need to build the software.

1. What problem are you trying to solve?

Answer: Water pipeline systems by using the AI stem in cities

2. Who faces this problem?

Answer: City people and water pipeline workers

3. What information/data would your system need?

Answer: The system needs information about the connections of pipes and sensors, previous damages, and where the water flow goes fast

4. Would you use AI, ML, DL, GenAI, or a combination? Why?

Answer: I use AI, ML, and DL to train the models by giving them pipeline connections, water flow, and temperature data. Sensors are used to know the water flow, temperature, and pressure of the water. GenAI is used to tell the workers where the problem is, give them suggestions to solve the problems, and generate images for clear explanations.

5. What would the system take as input?

Answer:It takes the inputs from the sensors.

6. What would the system produce as output?

Answer:The system would produce the location of the pipeline problem, the possible cause of the problem, and suggestions for the workers to solve it.

7. Give one example of how a user would interact with it.

Answer:A worker can check the system to know where the pipeline has a problem. If Area A has low water flow, the system shows that Area B may have a blockage and gives a suggestion to the worker.

8. What could go wrong with your solution?

Answers:The system may give wrong information if the sensor data is incorrect or if the AI and ML models were trained with old data.

9. How would you reduce or handle that risk?

Answers:We can regularly check the sensors and update the AI and ML models with new data. The workers can also verify the problem before repairing the pipeline

Example idea (do not copy): A college assistant that answers student questions about assignments, attendance rules, and course material.

## 8. Step 5 – Career Map: Where Can AI Take You?

Explore AI-related career paths. The goal is not to decide your career today; it is to understand what different roles actually do and what skills they require.

| Role                         | What do they do?                                                                           | Key skills                                                       | Typical tools/technologies                             | Would I explore it? Why?                                        |
|------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|
| AI / ML Engineer             | Builds and trains ML/AI models and integrates them into applications.                      | Python, ML algorithms, data preprocessing, APIs, problem-solving | Python, scikit-learn, TensorFlow/PyTorch, Git, APIs    | maybe— I want to explore how data is used to make predictions.  |
| Data Scientist               | Analyzes data, finds patterns, builds models, and gives predictions or insights.           | Python, statistics, ML, data analysis, visualization             | Python, Pandas, NumPy, scikit-learn, Jupyter, Power BI | maybe — I want to explore how data is used to make predictions. |
| Data Analyst                 | Analyzes data and creates reports and visualizations to help organizations make decisions. | Excel, SQL, data analysis, visualization, basic statistics       | Excel, SQL, Power BI, Tableau, Python                  | yes— it can help me improve my data-analysis skills.            |
| Generative AI / LLM Engineer | Builds applications using generative AI and large language                                 | Python, APIs, prompt engineering, LLMs,                          | Python, OpenAI APIs, Hugging Face, LangChain, vector   | yes— I am interested in how GenAI creates text,                 |

|                              |                                                                                             |                                                                       |                                                           |                                                                                    |
|------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|
|                              | models.                                                                                     | embeddings,                                                           | databases                                                 | images, and other content.                                                         |
| AI Product Manager           | Plans AI products and works with technical and business teams to build useful AI solutions. | Communication, product thinking, AI basics, problem-solving, teamwork | Jira, Notion, analytics tools, AI platforms               | maybe — I want to understand how AI products are planned and developed.            |
| AI Researcher                | Researches new AI methods, experiments with models, and develops new techniques.            | Mathematics, statistics, ML, deep learning, research skills           | Python, PyTorch, TensorFlow, Jupyter, research papers     | maybe— I want to explore AI research and understand how new models are developed.  |
| Computer Vision Engineer     | Builds AI systems that understand images and videos.                                        | Python, deep learning, image processing, computer vision              | OpenCV, PyTorch, TensorFlow, YOLO                         | yes — I am interested in image-related AI applications.                            |
| NLP Engineer                 | Builds AI systems that understand and process human language.                               | Python, NLP, ML, deep learning, language models                       | Python, Hugging Face, spaCy, PyTorch, LLM APIs            | yes — I want to explore how AI understands text and language.                      |
| MLOps / AI Platform Engineer | Deploys, monitors, and maintains ML models in production.                                   | Python, cloud, DevOps, APIs, Docker, CI/CD                            | Docker, Kubernetes, GitHub Actions, AWS/Azure/GCP, MLflow | maybe — I want to learn how trained models are deployed and maintained.            |
| AI Application Developer     | Develops applications that use AI models and APIs to provide AI-based features.             | Python/JavaScript, APIs, frontend/backend, databases, AI basics       | VS Code, Node.js, Python, APIs, MongoDB, Git              | yes — I am learning frontend and backend and want to connect AI with applications. |

## 9. Step 6 – Career Self-Check

Based on your exploration, answer:

Which AI-related role interests you most right now?

Answer:Generative AI / LLM Engineer

Why does it interest you?

Answer:It interests me because I want to learn how Generative AI can create new content and how LLMs understand and respond to user prompts.

What skills do you already have that may help?

Answers:Basics Python, prompt engineering.

What are 3 skills you need to learn next?

Answer:APIs, prompt engineering, LLMs, embeddings.

What would your 2-month learning roadmap look like?

Answer: Learn Python basics, Generative AI concepts, LLM basics, prompt engineering, and APIs. Learn how to use LLM APIs, build simple GenAI applications, work with embeddings and vector databases, and create one small GenAI project.

What would you like to build after learning the basics?

Answer: Generative AI / LLM Engineer

### 10. Step 7 – Common Errors: Find and Fix Them

For each statement below, decide whether it is correct, partially correct, or incorrect. Then explain why.

1. “AI and Machine Learning are exactly the same thing.”

Your verdict + correction: No, AI is a large field compared to ML. AI can respond without using ML models.

2. “Deep Learning is unrelated to Machine Learning.”

Your verdict + correction: No, DL is a subfield of ML, and they are related to each other.

3. “ChatGPT is just a search engine.”

Your verdict + correction: No, GPTs are not only search engines; they can also generate new data.

4. “Generative AI always gives correct answers.”

Your verdict + correction: No, sometimes AI can give wrong answers.

5. “If a company uses AI, it does not need humans anymore.”

Your verdict + correction: No, humans are required to analyze the responses and debug the code that AI generates, and for testing.

6. “More data automatically means a better AI system.”

Your verdict + correction: No, not always true.

### 11. Step 8 – Teach It Back

Pretend you are the trainer. Explain the following in a way that a Class 10 student can understand:

1. What is AI?:

AI is a field of computer science and engineering. It creates machines or software that can perform tasks like learning, problem-solving, and understanding like a human.

2. AI vs ML:

AI is a big field. It can perform tasks like learning, problem-solving, and understanding like a human. ML is a part of AI. It learns from a dataset and gives predictions.

### 3. ML vs DL:

ML is a part of AI. It learns from a dataset and gives predictions. DL is a subfield of ML. It has neural networks that help to analyze images, audio, and videos step by step, and it gives patterns as output.

### 4. What is Generative AI?:

GenAI is also a subfield of AI that can create new data. It uses large amounts of data from datasets. Based on that, it creates new data according to the prompt.

### 5. One real-world AI use case:

One real-world AI use case is Face Recognition Systems. When a face is captured by the camera, it analyzes the face, recognizes the person, and marks their attendance in the database.

### 6. One AI career path:

## 12. Final Challenge – 5-Minute AI Explanation

Record a 3–5 minute video or prepare a 1-page explanation as if you are teaching a friend. You must cover: AI → ML → DL → GenAI → real-world use case → career path.

Optional: Use a diagram, analogy, or story. Avoid reading a copied definition.

## 13. What to Submit

- AI explanation in your own words
- AI vs ML vs DL vs GenAI comparison table
- Relationship diagram
- 8-industry use-case table
- One AI solution design
- AI career map
- Career self-check
- Common errors and corrected explanations
- Teach-it-back answers
- Final 3–5 minute explanation OR 1-page summary

## 14. Self-Evaluation Checklist

| I can...                                               | Yes                        | Need More Practice         | Notes                                                                                |
|--------------------------------------------------------|----------------------------|----------------------------|--------------------------------------------------------------------------------------|
| Explain AI without using a memorized definition.       | <input type="checkbox"/> ✓ | <input type="checkbox"/>   | AI can perform tasks like learning, problem-solving, and understanding like a human. |
| Explain the relationship between AI, ML, DL and GenAI. | <input type="checkbox"/> ✓ | <input type="checkbox"/> ✓ | AI is a big field. ML and DL are used to learn from data and improve                 |

|                                                       |                                     |                                     |                                                                                                                                                                                                                            |
|-------------------------------------------------------|-------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |                                     |                                     | predictions. GenAI can create new data or content based on a prompt                                                                                                                                                        |
| Give real-world examples of AI.                       | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Face Recognition Systems, chatbots, YouTube recommendations, disease recognition, crop disease analysis, traffic prediction, and product recommendations.                                                                  |
| Identify whether a use case involves ML, DL or GenAI. | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | ML learns from data and gives predictions, DL analyzes complex data like images and speech, and GenAI creates new content.                                                                                                 |
| Explain at least 5 AI career roles.                   | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | AI/ML Engineer, Data Scientist, Data Analyst, Generative AI/LLM Engineer, AI Product Manager.                                                                                                                              |
| Design a basic AI solution for a real problem.        | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | For a water pipeline system, sensors collect water flow, temperature, and pressure data. AI identifies problems, ML predicts problems, DL finds complex patterns, and GenAI gives suggestions and explanations to workers. |
| Identify common AI misconceptions.                    | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | AI can give wrong analysis if the data is incorrect or old. We should not blindly trust AI because its predictions can sometimes be wrong.                                                                                 |
| Teach the basics of AI to another person.             | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | I can explain AI, ML, DL, and GenAI with simple real-world examples like face recognition, crop disease analysis, and water pipeline problems.                                                                             |

### 15. Rules for Using AI During This Assignment

- You MAY use ChatGPT, Gemini, Claude, Perplexity, YouTube, Google, and other learning resources.
- Use AI to explain, compare, question, and give examples—not to blindly copy your submission.
- If you use an AI-generated explanation, verify it using another source.
- Your final answers should be written in your own words.
- If you discover that an AI tool gave you an incorrect answer, mention the mistake and explain the correction. This is part of the learning.
- Add sources/links for important information you researched.

## 16. Trainer's Evaluation Guide

| Area                          | What to look for                                             | Suggested Weight |
|-------------------------------|--------------------------------------------------------------|------------------|
| Concept understanding         | Correct distinction between AI, ML, DL and GenAI             | 25%              |
| Own explanation               | Simple language, examples, analogies, not copied definitions | 20%              |
| Use cases                     | Relevant and correctly classified examples                   | 15%              |
| Practical thinking            | Problem → data → AI approach → output → risks                | 20%              |
| Career exploration            | Understanding of roles and skills                            | 10%              |
| Reflection & error correction | Can identify and fix misconceptions                          | 10%              |

**Remember:** The goal is not to know everything about AI today. The goal is to learn how to explore, question, understand, and explain AI.